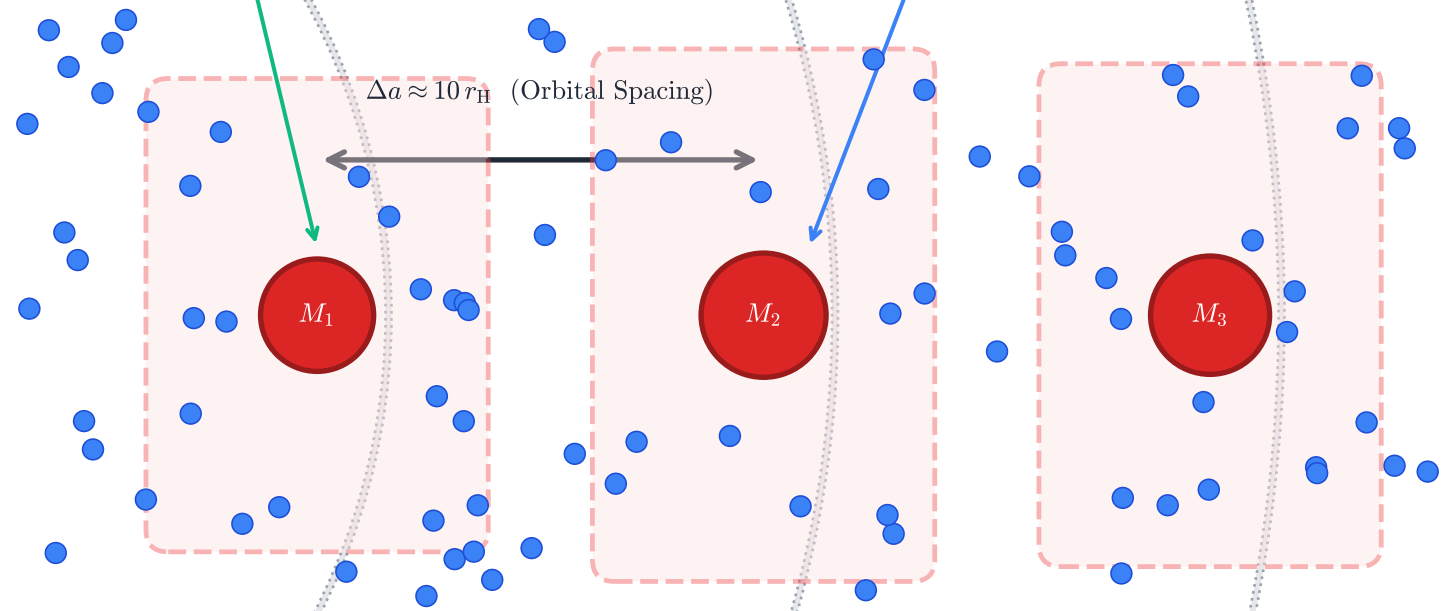


Dynamical Architecture of Oligarchic Planetary Growth (Kokubo & Ida 2000)



1. Viscous Stirring by Oligarchs
$$\left(\frac{de^2}{dt}\right)_{\text{VS}} \approx 25 \left(\frac{M}{M_*}\right)^{4/3} \frac{\Omega}{be^2} \ln \Lambda$$

2. Aerodynamic Gas Drag Damping
$$\left(\frac{de^2}{dt}\right)_{\text{drag}} \approx -\frac{C_D \rho_g v_K}{\rho_p v_p} e^3$$



3. Velocity Dispersion Equilibrium
$$\tilde{e}_{\text{eq}} \approx 5.0, \quad \tilde{i}_{\text{eq}} \approx 2.5 \quad (e \approx 2i)$$
$$\tilde{e}_{\text{eq}} \propto M^{1/15} m_0^{1/15} \Sigma_{\text{gas}}^{-1/5} a^{1/5}$$

4. Oligarchic Accretion & Equalization
$$\frac{dM}{dt} \approx 2\sqrt{2\pi} R^2 \left(\frac{v_{\text{esc}}}{cv_K}\right)^2 \Sigma_m \Omega \propto M^{2/3} \Sigma_m$$
$$\frac{1}{M} \frac{dM}{dt} \propto M^{-1/3} \rightarrow \text{Self-Regulated Mass Convergence}$$